

Killara HIGH SCHOOL

MATHEMATICS DEPARTMENT

2021

Year 12

HSC Mathematics Extension 1

TASK 2

Date issued: 26/02/20

Date due: 16/03/2020

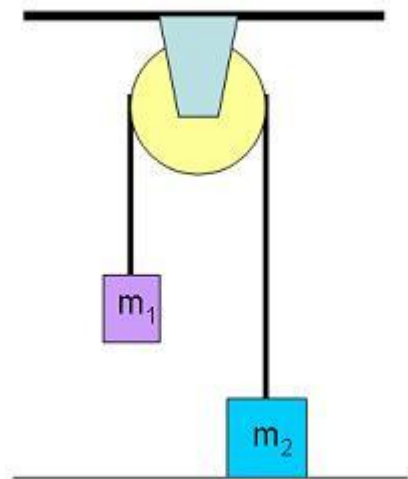
LEARNING OUTCOMES:

- Applies concepts and techniques involving vectors and projectiles to solve problems (ME12-2)
- Chooses and uses appropriate technology to solve problems in a range of contexts (ME12-6)
- Evaluates and justifies conclusions, communicating in a range of contexts (ME12-7)

Section I 40%	Part A – Modelling	/42
	Part B – Geometry	/23
	Part C – Modelling Orbits	/10
Section II 60%	45-minute examination	

- 1 Three boys are pulling a heavy trolley by means of three ropes. The boy in the middle is exerting a pull of 100N . The other boys, whose ropes both make an angle of 30° with the central rope, are pulling with forces of $50\sqrt{3}\text{N}$ and $100\sqrt{3}\text{N}$. Draw a diagram representing these forces and find the magnitude of the resultant pull on the trolley.

- 2 Two blocks with masses m_1 and m_2 are connected by a cord that passes over a massless and frictionless pulley.

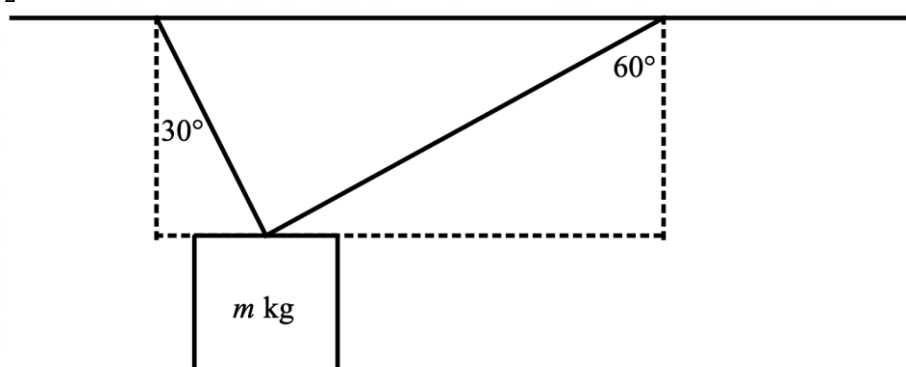


Find the mass of m_2 if $m_1=3\text{kg}$ and is accelerating upwards at 2.5ms^{-2}

3

[illegible]

- 3 A flowerpot of mass m kilograms is held in equilibrium by two light ropes, both of which are connected to a ceiling. The first rope makes an angle of 30° to the vertical and has tension T_1 newtons. The second rope makes an angle of 60° to the vertical and has a tension T_2 newtons.



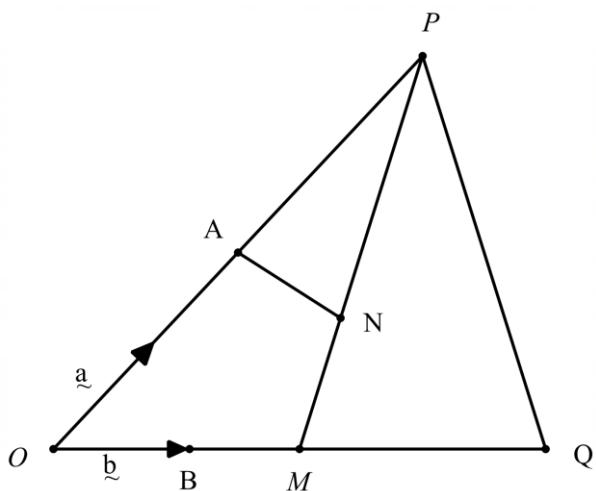
- (a) Show that $T_2 = \frac{T_1}{\sqrt{3}}$.

1

- (b) The first rope is strong, but the second rope will break if the tension in it exceeds 98N. Find the maximum value of m for which the flowerpot will remain in equilibrium.

3

- 4 In the diagram, $OA = \frac{1}{2}AP$ and $OB = \frac{1}{4}OQ$. M is the midpoint of OQ and $MN = \frac{1}{4}NP$.
 $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.



(a) Express, as simply as possible, in terms of $\mathbf{a}$ and/or $\mathbf{b}$.

(i) $\overrightarrow{OP}$ 1

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(ii) $\overrightarrow{OQ}$ 1

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(iii) $\overrightarrow{AQ}$ 1

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(iv) $\overrightarrow{MN}$ 1

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(v) $\overrightarrow{AN}$ 1

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- (b) Prove that AN , when produced, will pass through Q . 1

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- (c) Find the ratio $AN:NQ$ 1

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- (d) Calculate the value of $\frac{\text{Area of } \triangle OPQ}{\text{Area of } \triangle PMQ}$ 2

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This is the end of the examination.

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Answers

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Section I 40%	Part A – Modelling	/42
	Part B – Geometry	/23
	Part C – Modelling Orbits	/10
Section II 60%	45-minute examination	

*↳ 35 – 40 mins
(Last question!).*

- 1 Three boys are pulling a heavy trolley by means of three ropes. The boy in the middle is exerting a pull of 100N. The other boys, whose ropes both make an angle of 30° with the central rope, are pulling with forces of $50\sqrt{3}$ N and $100\sqrt{3}$ N. Draw a diagram representing these forces and find the magnitude of the resultant pull on the trolley.

① diagram

$\frac{1}{2}$
 $\frac{\sqrt{3}}{2}$
 $\frac{50\sqrt{3}}{2}$

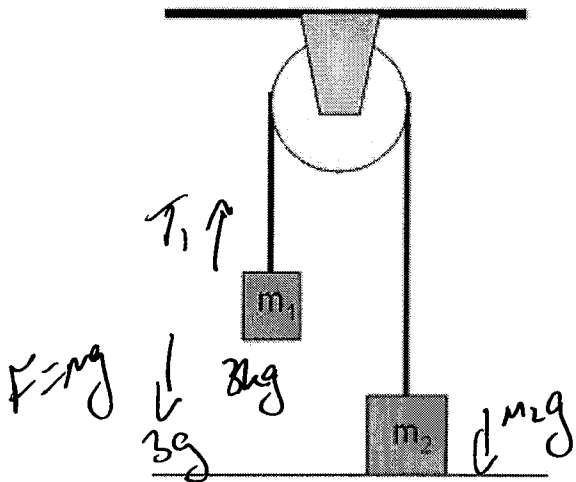
$\uparrow \Rightarrow 100 + \frac{50\sqrt{3} \times \sqrt{3}}{2} + \frac{100\sqrt{3} \times \sqrt{3}}{2} = 325$
 ① for vertical (cos)

$\Rightarrow \Rightarrow 100\sqrt{3} \times \frac{1}{2} - 50\sqrt{3} \times \frac{1}{2} = 25\sqrt{3}$
 ① for horizontal (sin)

$\sqrt{325^2 + (25\sqrt{3})^2}$
 $= 50\sqrt{43}$ N
 $= 327.87$ N

① for resultant

- 2 Two blocks with masses m_1 and m_2 are connected by a cord that passes over a massless and frictionless pulley.



① for $F = mg = ma$.

& some work on diagram

Find the mass of m_2 if $m_1 = 3\text{kg}$ and is accelerating upwards at 2.5ms^{-2}

3

~~$F = T - 3g$~~

~~$F = M a$~~
 ~~$F = M_2 g - 3g$~~

~~$F = M a$~~
 ~~2.5×3~~

~~$F_R = 7.5$~~

$F_2 = W_2 - T = m_2 a$

~~$F_2 = M a$~~

$W_2 - T = m_2 a$

$m_2 g - 36.9 = m_2 \times 2.5$

$m_2 (9 - 2.5) = 36.9$

$m_2 = 5.05 \text{ kg}$

$F_1 = T - W_1 = m_1 a$

$T - 3g = 2.5 \times 3$

$T = 7.5 + 3g$

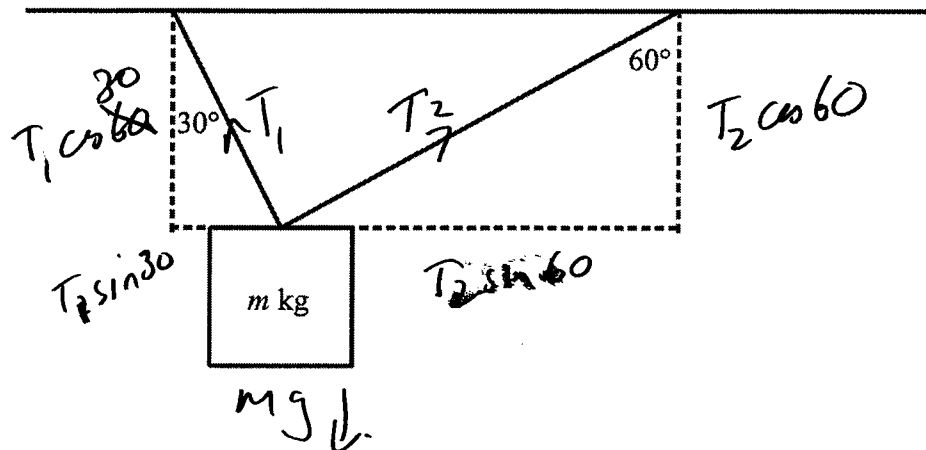
$= 36.9$

① for T

① for m_2

37.5
 $W/10$

- 3 A flowerpot of mass m kilograms is held in equilibrium by two light ropes, both of which are connected to a ceiling. The first rope makes an angle of 30° to the vertical and has tension T_1 newtons. The second rope makes an angle of 60° to the vertical and has a tension T_2 newtons.



- (a) Show that $T_2 = \frac{T_1}{\sqrt{3}}$.

$$T_1 \sin 30 = T_2 \sin 60$$

$$T_1/2 = T_2 \times \frac{\sqrt{3}}{2}$$

$$\sqrt{3} T_2 = T_1$$

$$T_2 = \frac{T_1}{\sqrt{3}}$$

(1)

- (b) The first rope is strong, but the second rope will break if the tension in it exceeds 98N. Find the maximum value of m for which the flowerpot will remain in equilibrium.

$$T_1 \cos 30 + T_2 \cos 60 = mg \quad (1) \text{ for equilibrium}$$

$$\frac{\sqrt{3} T_1}{2} + \frac{T_2}{2} = mg$$

$$\sqrt{3} T_1 + T_2 = 2mg$$

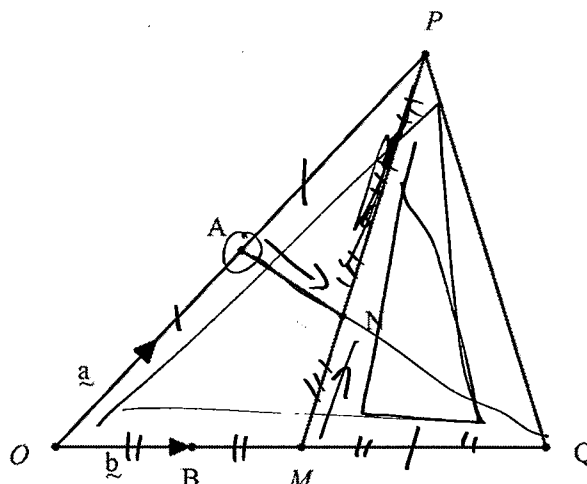
$$3T_2 + T_2 = 2mg \quad (1) \text{ for substitution}$$

$$4T_2 = 2mg$$

$$2 \times 98 = mg$$

$$m = 20 \text{ kg} \quad (1) \text{ for result}$$

- 4 In the diagram, $OA = \frac{1}{2}AP$ and $OB = \frac{1}{4}OQ$. M is the midpoint of OQ and $MN = \frac{1}{4}NP$.
 $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.



(a) Express, as simply as possible, in terms of $\mathbf{a}$ and/or $\mathbf{b}$.

(i) $\overrightarrow{OP}$

$$\cancel{2\mathbf{a}} \quad 3\mathbf{a}$$

①

(ii) $\overrightarrow{OQ}$

$$4\mathbf{b}$$

①

(iii) $\overrightarrow{AQ}$

$$4\mathbf{b} - \mathbf{a}$$

①

(iv) $\overrightarrow{MN}$

$$\overrightarrow{MP} = \cancel{2\mathbf{b}} \quad 3\mathbf{a} - 2\mathbf{b}$$

$$\overrightarrow{MN} = \frac{3\mathbf{a} - 2\mathbf{b}}{5}$$

①

(v) $\overrightarrow{AN}$

$$\frac{2\mathbf{b} + 3\mathbf{a} - 2\mathbf{b}}{5} - \mathbf{a}$$

$$\frac{10\mathbf{b} + 3\mathbf{a} - 2\mathbf{b} - 5\mathbf{a}}{5} = \frac{8\mathbf{b}}{5} - \frac{2\mathbf{a}}{5}$$

①